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Arabic chatbot technologies: A scoping review

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ABSTRACT

Background: Chatbots have been widely used in many spheres of life from customer services to mental health companions. Despite the breakthroughs in achieving human-like conversations, Arabic language chatbots driven by AI and NLP are relatively scarce due to the complex nature of the Arabic language.
Objective: We aim to review published literature on Arabic chatbots to gain insight into the technologies used highlighting the gap in this emerging field.
Methods: To identify relevant studies, we searched eight bibliographic databases and conducted backward and forward reference checking. Two reviewers independently performed study selection and data extraction. The extracted data was synthesized using a narrative approach.
Results: We included 18 of 1755 retrieved publications. Thirteen unique chatbots were identified from the 18 studies. ArabChat was the most common chatbot in the included studies ($n = 5$). The type of Arabic language in most chatbots ($n = 13$) was Modern Standard Arabic. The input and output modalities used in 17 chatbots were only text. Most chatbots ($n = 14$) were able to have long conversations. The majority of the chatbots ($n = 14$) were developed to serve a specific purpose (Closed domain). A retrieval-based model was used for developing most chatbots ($n = 17$).
Conclusion: Despite a large number of chatbots worldwide, there is relatively a small number of Arabic language chatbots. Furthermore, the available Arabic language chatbots are less advanced than other language chatbots. Researchers should develop more Arabic language chatbots that are based on more advanced input and output modalities, generative-based models, and natural language processing (NLP).

1. Introduction

1.1. Background

Chatbots are computer programs that automatically communicate with users via text or spoken format. Eliza was the first chatbot, which was developed by the Massachusetts Institute of Technology (MIT) in 1966 [1]. Since then, chatbots have been used in various fields such as healthcare, business, education, and banking. With the sudden changes in work practices and decrease in face-to-face services due to the COVID-19 pandemic, we have seen a rapid increase in the usage of chatbots.

Chatbots consist of three key components: interface, conversation engine, and knowledge base. The interface can be in the form of a webpage, instant messaging app (e.g., WhatsApp and telegram), and a dedicated mobile application. The conversation engine manages the semantic context of the conversation. The knowledge base is the brain and content of the conversation. According to the knowledge base, there are 3 types of chatbots: rule-based, retrieval-based, and generative-based. Rule-based chatbots used a fixed set of rules (decision tree models) to interact with the user. This type of chatbot used only predefined responses to reply to the users' questions and cannot learn through user interactions. Retrieval-based chatbots also have predefined responses but allow some heuristics to extract these responses as a means of shortcut to the complete decision tree approach. This type of chatbot does not generate new responses although it can contain elements of AI. An example of retrieval-based chatbots is Mitsuku, which constructs songs and poems based on its knowledge base of over 3000

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objects and predefined responses of 300,000 [2]. The generative-based chatbots can create a new dialogue using large amounts of conversational data. The generative-based model can use a combination of techniques including supervised learning, unsupervised learning, reinforcement learning, and adversarial learning.

Chatbots are also able to have either short conversations (a single response is produced for a single Q&A conversation) or long conversations (large amounts of information exchanged throughout conversations). The conversation domain of chatbots can be closed (i.e., aim to serve a specific purpose or topic; the knowledge required for output response is limited) or open (aim to serve different types of conversation on different topics).

One of the most important challenges faced by chatbot developers is allowing the user to feel as though they are speaking to a real person. Applying Artificial Intelligence (AI) algorithm has allowed a breakthrough in achieving human-like responses. AI algorithms require the processing of large amounts of data and interactions. Such amount of data has been somewhat achieved with AI-powered conversational agents such as Facebook messenger bots, Alexa, Microsoft Cortana, and Siri [3]. These datasets and other openly available corpora remain largely in the English language. This means that even those who have a good grasp of the algorithms often lack the data to develop and further optimize Arabic language chatbots. A previous study looked into the freely available Arabic corpora and found only 66 sources. However, not all of these would be useful for building AI-based chatbots [4]. Another challenge of developing Arabic language chatbots is that the Arabic language has rich morphology, orthographic variations, ambiguity, and multiple dialects.

1.2. Research problem and aim

While there is a large number of chatbots worldwide, chatbots that use the Arabic language to communicate with users are somewhat scarce. In 2016, the platforms of Pandorabots, Facebook, Microsoft, and Kik were used to develop 285,000, 11,000, 30,000, and 6000 chatbots, respectively [5].

A review of the literature found 56 studies about chatbots [6]. However, this may not be a true representation of all the literature available as it had limited inclusion criteria and searched only Google Scholar. Another review found only 14 studies about Arabic language chatbots [7]. Due to the low numbers of chatbots available for the Arabic language, little is reported on the methodologies used, thereby, this may hinder the progress for those seeking to develop chatbots in the Arabic language. Therefore, we aimed to explore the features of Arabic language chatbots that have been reported in the literature.

2. Methods

To achieve the objective of the paper, we carried out a scoping review based on the guidelines of PRISMA Extension for Scoping Reviews (PRISMA-ScR) [8]. The methods used in this review are detailed in the following sections.

2.1. Search strategy

2.1.1. Search sources

The retrieved studies were collected from eight online databases: ACM Digital Library, IEEE Xplore, Springer, Google Scholar, ScienceDirect, Embase, Medline, and PsychINFO. For Google Scholar, only the first 100 studies were checked as a large number of search results returned. Furthermore, we used forward and backward reference list checking. The search process for this review started on the 20th of December 2020 and progressed till the 25th of December 2020.

2.1.2. Search terms

The search terms used in this review are related to the Arabic language and chatbots. Specifically, the search query used in this review is as follows: (("conversational agent*" OR "conversational bot*" OR "conversational system*" OR "conversational interface*" OR "chatbot*" OR "chat bot*" OR "chatterbot*" OR "chatter bot*" OR "smartbot*" OR "smart bot*" OR "smart-bot*" OR "ArabChat*" OR "Chat agent" OR "Interactive agent" OR "artificial conversational")). Medical Subject Headings were used in three databases: MEDLINE, EMBASE, and PsychINFO.

2.2. Study eligibility criteria

We included studies describing chatbots that used the Arabic language to interact with users regardless of their types, platforms, input and output modalities, and domain of conversation. However, we excluded chatbots that are controlled by a human operator. We focused on peer-reviewed journal articles, reports, conference papers, book chapters, and white papers that were published between 2014 and 2020. On the other hand, we excluded conference abstracts and posters, magazines, and newspapers. Studies written in languages other than English were not considered in this review.

2.3. Study selection

The filtering process involved three stages: removing duplicates, screening titles and abstracts, and checking full texts. Two reviewers individually performed the study selection process, and any dissimilarities were solved by discussion between the reviewers. Rayyan software was used for the study selection process [9].

2.4. Data extraction and synthesis

We excluded data related to study meta-data (i.e., authors name, year of publication, and country of publication) and features of Arabic chatbots (i.e., their name, domain, input and output modality, dataset model, conversation length). A narrative approach was used to summarize the extracted data. Data extracted from each study are shown in Appendix A.

3. Results

3.1. Search results

As presented in Fig. 1, 1755 citations were identified after searching eight databases. Ninety-nine duplicates were removed. After reading the titles and abstracts of the remaining 1656 citations, we removed1636 citations. We went through the full texts of the remaining 20 articles and excluded 5 studies. Three additional studies were identified by checking the studies that cited the included studies. This review included a total of 18 studies.

3.2. Characteristics of the included studies

The included articles were published between 2004 and 2020 (Table 1). Interestingly, no studies on Arabic language chatbots were published from 2005 to 2010. The included studies were conducted in 8 different studies. The countries that published the largest number of studies were the United Kingdom (UK) and Jordan (5 studies each). More than half (10/18) of the papers included were conference papers. Table 2

3.3. Features of the Arabic chatbots

There were 13 unique Arabic chatbots in the included studies. The Arabic language chatbots were given a specific name in 12 studies. ArabChat was the most common chatbot in the included studies ($n = 5$). The type of Arabic language used in chatbots was Modern Standard

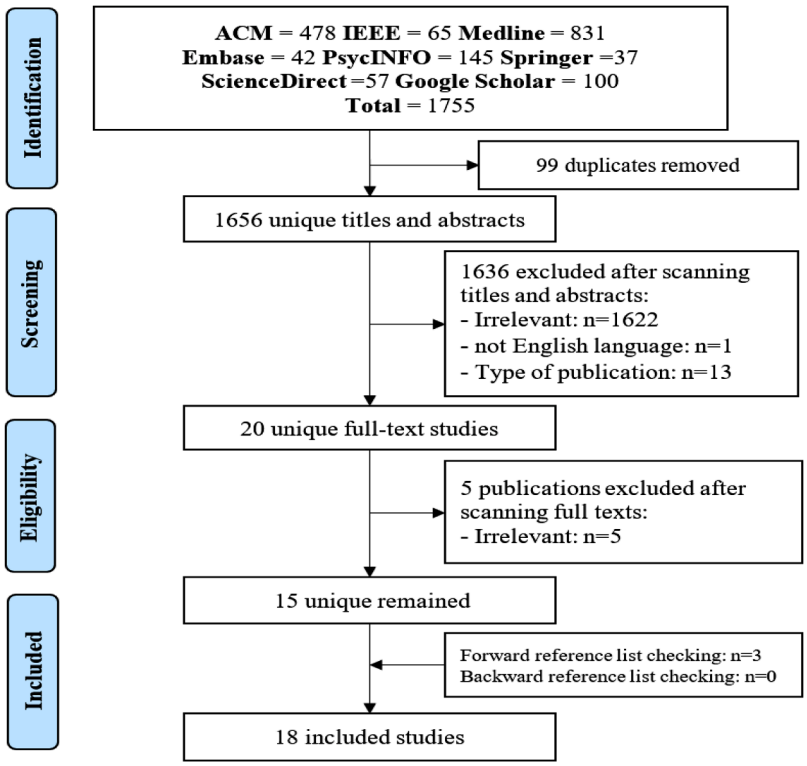

Fig. 1. Flow chart of study selection process.

Table 1
Characteristics of the included studies.

Characteristics	Number of studies	Studies
Year	2004: 2	[10,11]
	2011: 1	[12]
	2013: 2	[13,14]
	2014: 1	[15]
	2015: 2	[16,17]
	2016: 2	[18,19]
	2017: 1	[20]
	2018: 2	[21,22]
	2019: 3	[23–25]
	2020: 2	[26,27]
Country	UAE: 1	[19]
	Saudi Arabia: 3	[22,25,27]
	Italy: 1	[23]
	Lebanon: 1	[26]
	Jordan: 5	[12,14,16,17,18]
	UK: 5	[10,11,13,15,20]
	Syria: 1	[24]
	Malaysia: 1	[21]
Publication type	Conference paper: 10	[10,11,13–15,19–21,23,24]
	Journal article: 7	[12,16–18,22,25,27]
	Workshop paper: 1	[26]

Table 2
Chatbot descriptions.

Characteristics	Number of studies	Studies
Name of chatbot	Yes: 12	[13–21,23,25,27]
	No: 6	[10–12,22,24,26]
Arabic language	Classical Arabic: 2	[10,11]
	Modern Standard Arabic: 13	[12–18,20–23,25,26]
	Dialectical Arabic: 3	[19,24,27]
Input modality[1]	Text: 18	[10–27]
	Speech: 1	[25]
Output modality[2]	Text: 18	[10–27]
	Speech: 1	[25]
Conversation length	Long: 14	[13–21,23–27]
	Short: 4	[10–12,22]
Domain	Open: 4	[14,16,19,26]
	Closed: 14	[10–13,15,17,18,20–25, 27]
Dataset model	Retrieval- Based: 17	[10–25,27]
	Generative- Based: 1	[26]
NLP used	Yes: 8	[10,11,14,19,23–26]
	No: 10	[12,13,15–18,20–22,27]

1: number of studies does not add up as one study used 2 input modalities.
2: number of studies does not add up as one study used 2 output modalities.

Arabic in 13 studies, local Arabic dialects (Egyptian, Saudi, and Levantine) in 3 studies, and classical Arabic (Quranic Arabic) in 2 studies.

The input and output modalities used in chatbots were only text in 17 studies and text and speech in 1 study. Most chatbots ($n = 14$) were able to have long conversations. The majority of the chatbots ($n = 14$) were developed to serve a specific purpose (Closed domain) such as health-care services, restaurant orders, airline ticketing, and education. Retrieval-based chatbots were the most common type of chatbots in the included studies ($n = 17$). In 8 studies, Natural language processing (NLP) techniques were used in chatbots.

4. Discussion

4.1. Principal findings

In this scoping review, we report on the literature based on Arabic chatbots and included 18 publications. The main finding of this scoping review is the lack of publications on Arabic chatbots. Despite including a wider range of years in the inclusion criteria, and searching more databases than previous similar studies, the number of papers was limited. When we searched app stores, we found a few additional Arabic chatbots (e.g., Rammas, Al-HajBot, Theyabi). This scoping review observed a clear gap in the literature.

Classical Arabic was the most widely used language amongst the

included chatbots. Written Arabic text is classified into three categories: (i) Classical Arabic (ii) Modern Standard Arabic and (iii) Dialectical Arabic [7]. Quranic Arabic is often categorized under classical Arabic, for the purpose of this study we marked it as such. The MSA is the formal official language used in academia, law, and newspaper, however, it is not spoken as a mother tongue in the Arab world. On the other hand, the DA is the mother language of Arabs and it will vary from one region to another and country to country. Traditionally, DA rarely exists in written form as it was a mostly spoken variety, however in recent years with the appearance of social media, DA became the main Arabic variety used online.

Fourteen of the included chatbots used CA. Two of the earliest studies included in this review are chatbots that have adapted Quran as their training corpus [10,11]. One study accepts an English word as an input and returns a verse from the Quran in both English and Arabic [10]. The other study accepts an Arabic word and replies with an appropriate verse from the Quran [11]. Three unique chatbots had used three different dialects in their chatbots: Egyptian, Saudi, and Levantine Arabic dialects. In addition, two of the chatbots also had the option of English language. Botta is the first Arabic dialect chatbot developed in 2016 that simulates conversations in the Egyptian dialect [19].

The majority of the chatbots were based on Pattern Matching techniques while only seven of the included chatbots had Natural language programming techniques in them. Pattern matching is not expensive computationally and is easy to understand and has simple preprocessing steps [17]. Out of the seven NLP-based chatbots included, two were adapted from ALICE [10,11] and NLP for another chatbot was offered by IBM Watson conversation [23]. This again shows the lack of more complex technologies in the Arabic language. Arabic NLP is progressing slowly compared to other languages due to the complexity of the language. NLP techniques in Arabic chatbots are even rarer.

4.2. Strengths and limitations

4.2.1. Strengths

To the best of our knowledge, this is the first scoping review that was conducted based on PRISMA Extension and focused on Arabic chatbots. Hence, it provides a clear understanding for readers and researchers who are working in this field, pointing out the lack of Arabic chatbots and how more research is needed in this area. We were extensive having retrieved relevant studies by searching healthcare and information technology online databases. We minimized the publication bias by conducting backward and forward reference list checking. In addition, two independent reviewers were assigned for study selection and data extraction to avoid selection bias. This review aimed to cover all studies that were published between 2004 and 2020.

4.2.2. Limitations

This review excluded Arabic chatbots embedded in robotics, serious games, SMS, and telephone, as well as those controlled by human operators. Hence, this review did not dive deep into the techniques used to develop these chatbots because it does not serve the purpose of the scoping review. The search was restricted to English studies published between 2004 and 2020 and we could not cover some of the online databases (e.g., Web of Science, Scopus, ERIC, DOAJ, and JSTOR), therefore, we may have missed some relevant articles.

4.3. Practical and research implications

This review can be used as a reference for researchers interested in exploring Arabic chatbots by having easy access to the list of the publications of Arabic chatbots with their characteristics such as the language variety used, the input and output modality, the domain used, the dataset model and also the technology used, for instance out of the 18, only eight studies have used NLP techniques.

The findings of this review will help direct future research related to Arabic chatbots as it identified the gaps in the literature. For example, as most of the identified studies were based on Pattern Matching techniques while a limited number of studies were based on NLP techniques and other advanced technologies, this will eventually encourage more work based on NLP and other advanced AI technologies in future studies given that NLP is an established AI technology used with advanced conversational agents.

Another important aspect identified in this survey is the lack of focus on DA as only 3 studies out of 18 covered DA. Since in most cases the goal of the chatbots is to talk to people in their language, it is crucial that future studies focus more on DA when building their models.

We foresee an increase in the development and use of chatbots in various fields from customer services to mental health by combining existing fast-evolving technologies such as Machine Learning, AI, and NLP with evidence-based partnerships. For example, between developers and fields such as mental health professionals. Following the COVID-19 epidemic and its influence on global mental health and wellbeing, particularly for stress, anxiety, and depression, their involvement in such care is projected to grow. Chatbots offer a genuine benefit and efficient means for the business sector but in the current climate more importantly in psychiatric conversational therapies, especially in areas of the world where the gold standard one-on-one psychiatrist-patient discussions are just not practical or economical. In parts of the Arab speaking world such as the Gulf region speaking out about mental health is still a sensitive and taboo subject, having an intelligent AI driven evidence based virtual chatbot to speak to would provide access to mental health advice where culturally people are reluctant to open up to psychiatrists and other experts. A greater effort is needed by Arab language experts, software engineers including app developers, researchers and health care professionals to bring the availability of quality useable apps to the standard of those apps aimed at English speakers.

5. Conclusion

We presented the detailed results of our review on Chatbots and Conversational agents targeting the Arabic languages and related varieties. Despite some recent advances in chatbot development, the Arabic language is still lagging behind when compared to other languages such as English. The 18 studies revealed that the existing Arabic Chabot projects were in most cases not targeting the main spoken Arabic dialectal varieties and, Arabic language chatbots are less advanced than other language chatbots when it comes to application of AI technologies. Focus by developers and researchers is needed to develop more Arabic language chatbots that are based on more advanced input and output modalities, generative-based models, and natural language processing (NLP). There is a clear gap that we hope to be addressed in the future, especially with the growing need for human-like chatbots. There is a real gap and missed potential targeting the Arab speaking world especially given the current COVID-19 situation in fields such as mental health.

CRediT authorship contribution statement

Arfan Ahmed: Writing – original draft. **Nashva Ali:** Data curation. **Mahmood Alzubaidi:** Methodology. **Wajdi Zaghouani:** Supervision. **Alaa Abd-alrazaq:** Supervision. **Mowafa Househ:** Project administration.

Declaration of Competing Interest

The authors have no competing interests to declare.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.cmpbup.2022.100057.

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